Week 4 Worksheet

Group members: _____

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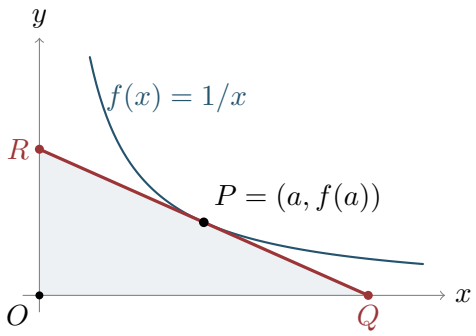
Problem 1. A tangent line and a limiting area Monday: Sections 3.1–3.2

Let $f(x) = 1/x$ for $x > 0$. For $a > 0$, let $P = (a, f(a))$ lie on its graph. The tangent line at P meets the positive x-axis at Q and the positive y-axis at R. Together with the origin O, these intercepts form a triangle.



Let $A(a)$ be the area of $\triangle OQR$. Our goal is to determine $\lim_{a\to 0^+} A(a)$.

(a) Before computing, predict what happens to $A(a)$ as $a \to 0^+$. Sketch a second possible position of the tangent line for a smaller positive value of a. Record your group's prediction and geometric reasoning.

(b) Use the **definition of the derivative** to find $f'(a)$, the slope of the tangent at P. Show the simplification of the difference quotient before taking its limit. Do not use a differentiation rule here.

Hint: Start with $\dfrac{\frac{1}{a+h} - \frac{1}{a}}{h}$.

Problem 1, continued.

(c) Use your result from (b) to write the tangent line at P as a function $\ell(x)$. Then find the intercepts Q and R in terms of a.

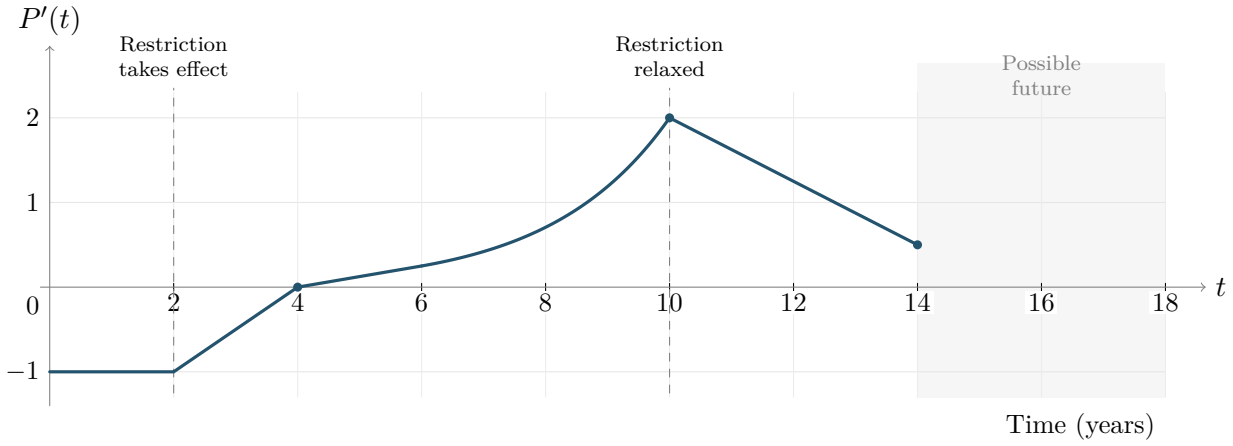
(d) Find a formula for $A(a)$ and evaluate $\lim_{a\to 0^+} A(a)$. Compare the result with your prediction. Explain how the changing base and height of the triangle account for your answer.

(e) Let $L(a)$ be the perimeter of $\triangle OQR$. Find a formula for $L(a)$ and evaluate $\lim_{a\to 0^+} L(a)$. Compare the limiting behavior of the perimeter with that of the area. How can both results hold?

(f) Could a physical triangle realize the limiting construction as $a \to 0^+$? Explain why or why not using your results for area and perimeter. Distinguish the limit from a triangle with a very small positive value of a.

Problem 2. Is the fish population recovering? Wednesday: Sections 3.3–3.4

Let $P(t)$ be the fish population in a lake, measured in thousands of fish, t years after a study begins. Initially, $P(0) = 20$. In this fictional model, a fishing restriction takes effect at $t = 2$ and is relaxed at $t = 10$. The graph shows $P'(t)$, measured in thousands of fish per year, for $0 \le t \le 14$.



(a) Interpret $P'(1) = -1$ and $P'(10) = 2$ in the context of the lake. Include units. Does either value tell you how many fish are in the lake?

(b) When is the population decreasing, and when is it increasing? When is it smallest during the first 14 years? Does the decline stop as soon as the restriction takes effect? Support your answers with the graph.

(c) For each of the following intervals, state whether the population is increasing or decreasing and whether its rate of change is increasing or decreasing. For the two middle intervals, also compare how quickly the growth rate increases.

$2 < t < 4$:

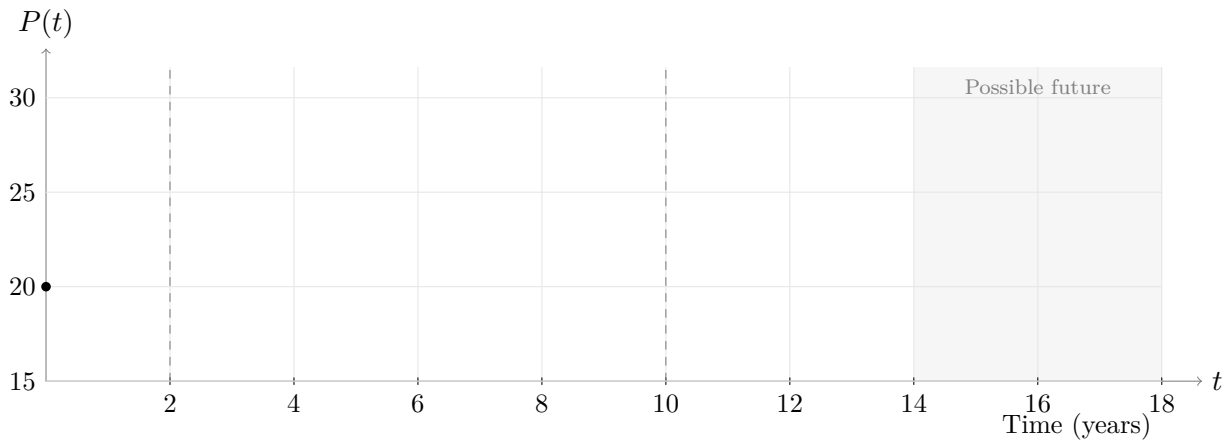
$4 < t < 6$:

$6 < t < 10$:

$10 < t < 14$:

Problem 2, continued.

(d) Sketch $P(t)$ for $0 \le t \le 14$, starting at $P(0) = 20$. Show where the graph falls, rises, has a horizontal tangent, and rises most steeply. Focus on the shape; exact later population values are not required.



(e) A lake manager says, "After the restriction was relaxed, the graph went down, so we began losing fish again." Is this conclusion supported by the graph? Explain what would have to happen to $P'(t)$ for the population to begin declining again.

(f) Suppose officials change the fishing restrictions at $t = 14$. Choose one policy: **strengthen the restrictions** or **remove the remaining restrictions**. Describe how you expect the population's rate of change to respond, including any delay. In the shaded regions, use dashed curves to extend both $P'(t)$ and $P(t)$ through $t = 18$. Make sure the slopes of your population sketch agree with your predicted rate of change.